Unified regularization method using selection probability to identify pleiotropic variants associated with mixed-type outcomes

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Introduction

- Although genome-wide association studies(GWAS) have successfully identified numerous loci at which common variants have an influence on a disease or a trait, a mount of heritability still remains unexplained.
- The pleiotropy, which refers to the phenomenon where a variant or a genetic region affects multiple traits at a time, is one of the key factors to understand the missing heritability, because we can more correctly identify outcome-related variants, which cannot be selected in single-phenotype analysis, by making full use of multiple phenotype outcomes. Additionally, it could enable us to prevent multiple diseases simultaneously with single pleiotropic variant.
- Although there are a variety of phenotypes in reality, it is difficult to analyze the mixed-type phenotypes due to the lack of available statistical distributions.
- To this end, there have been many studies to develop a new statistical methods to identify cross-phenotype variants associated with mixed-type outcomes, but most of them proposed recently are limited to the meta-analysis which utilizes the summary statistics across multiple phenotype outcomes. However, they cannot accommodate the correlations among variants which plays an important role in an analysis of genomic data.
- In this work, we proposed a unified selection approach that combines the univariate regularization results into a selection probability that does not depend on tuning parameter. Since our proposed method utilizes the individual-level genomic data, it is more powerful than the existing meta-analyses.

Methods

- Assume that we observed p genotypes of SNP and q phenotype outcomes that could be continuous, binary or categorical from n individuals.
- **Multivariate Generalized Linear Regression Model**

$$g_k(\mu_{ik}) = \boldsymbol{x}_i^T \boldsymbol{\beta}_k, \quad \text{and} \quad \mu_{ik} = E(y_{ik}|\boldsymbol{x}_i) \tag{1}$$

where $\boldsymbol{x}_i = (x_{i1}, \cdots, x_{ip})^T \in \mathbb{R}^p$, $\boldsymbol{y}_k = (y_{1k}, \cdots, y_{nk})^T \in \mathbb{R}^n$, $k = 1, \cdots, q$, $\boldsymbol{\beta}_k = (\beta_{1k}, \cdots, \beta_{pk})^T \in \mathbb{R}^p$ and $g_k(\cdot)$ is a link function corresponding to the k-th response. Without loss of generality, we assume that all predictors have mean zero, i.e., $\sum_{i=1}^n x_{ij} = 0$ for all $j = 1, \cdots, p$. Each link function will vary with each response type such as continuous, binary and categorical.

- Univariate **regularization model** can estimate the regression coefficients even in high-dimensional data as follows:

$$\underset{\boldsymbol{\beta}_k}{\arg\min} - \ell_k(\boldsymbol{y}_k, \boldsymbol{X}\boldsymbol{\beta}_k) + P_\Lambda(\boldsymbol{\beta}_k), \tag{2}$$

where $\boldsymbol{X} = (\boldsymbol{x}_1, \cdots, \boldsymbol{x}_n)^T$, ℓ_k is a log-likelihood function corresponding to the link function g_k, P_Λ is a penalty function, Λ is a set of tuning parameters.

- **The elastic-net penalty function**:

$$P_{\alpha,\lambda}(\boldsymbol{\beta}_k) = \lambda\left((1-\alpha)\|\boldsymbol{\beta}_k\|_2^2/2 + \alpha\|\boldsymbol{\beta}_k\|_1\right)$$

where (1) λ is a penalty parameter that governs how much panelty will be imposed on the parameter estimator $\boldsymbol{\beta}_k$, (2) α is a mixing proportion between two penalty functions, and (3) $\|\cdot\|_1$ and $\|\cdot\|_2$ are $\mathcal{L}_1$ and $\mathcal{L}_2$ norm.

- **Research problem** when unifying the multiple regularization models.
  1. A causal SNP could have different strengths of association for each of multiple outcomes, where the unified approach as above cannot detect SNPs which have relatively weak associations on a few of outcomes.
     → Degrees of freedom as a tuning parameter
  2. How to make full use of all phenotypic informations rather than a single phenotype in order to improve the statistical power.
     → Aggregation of the selected associations to be associated with a variant.

- To address the research problem, we adopted the tuning paramter η number of selected variables as an estimator of the degrees of freedom for the selection model instead of penalty parameter λ and the aggregation approach to take a summation of the selected associations across multiple outcomes.
- **An algorithm of unified elastic-net model with df-aggregation approach (UNET-df)**

(1) Set a grid of tuning parameters: $\eta \in \mathbb{N}$ and fixed $\alpha = 0.1$.
(2) Draw a random subsample I_h with size of $[n/2]$ without replacement.
(3) Estimate regression coefficients $\beta_{jk}^{\alpha,\eta}(I_h)$ for a high-dimensional data using elastic-net penalty for a given α and η.
(4) Our proposed selection probability can be written as:

$$\hat{\Pi}_j = \max_{\alpha,\eta}\left[\frac{1}{H}\sum_{h=1}^{H}\sum_{k=1}^{q}\mathbb{I}\left(\hat{\beta}_{jk}^{\alpha,\eta}(I_h) \neq 0\right)/C\right],$$

where $C = \max_j \frac{1}{H}\sum_{h=1}^{H}\sum_{k=1}^{q}\mathbb{I}\left(\hat{\beta}_{jk}^{\alpha,\eta}(I_h) \neq 0\right)$ is a normalizing constant which enables us to scale the selection probabilities across a grid of tuning parameters.

Simulation studies

- Data generation process:
  1. [Design matrices] We first generate two p-dimensional vector $(x_{i1}^1, \cdots, x_{ip}^1)^T$ and $(x_{i1}^2, \cdots, x_{ip}^2)^T$ from a multivariate normal distribution $\mathcal{N}_q(\boldsymbol{0}, \Sigma_x)$, where Σ_x is the auto-regressive covariance matrix whose (u, v)-th element is $0.95^{|u-v|}$.
  2. [Minor Allele Frequencies] Next, we generated a p-dimensional vector $\boldsymbol{f} = (f_1, \cdots, f_p)^T$ of minor allele frequency (MAF) from an uniform distribution $\mathcal{U}(0.05, 0.5)$ similarly as in Shi et al.(2019).
  3. [Genotype] We can then generate the simulated SNP data by discretizing each of p vectors to genotype code {0, 1, 2} as:

$$x_{ij} = \mathbb{I}(q_{ij}^1 < f_j) + \mathbb{I}(q_{ij}^2 < f_j), \tag{3}$$

where $q_{ij}^1 = \frac{1}{n}\sum_{i'=1}^n \mathbb{I}(x_{ij}^1 \leq x_{i'j}^1)$ and $q_{ij}^2 = \frac{1}{n}\sum_{i'=1}^n \mathbb{I}(x_{ij}^2 \leq x_{i'j}^2)$ are the reverse quantile of x_{ij}^1 and x_{ij}^2 for all $j = 1, \cdots, p$, respectively.

  4. [Phenotype] To construct the mixed-type outcomes, we first considered continuous and binary phenotypes. If we denote by $\boldsymbol{y}_i$ the i-th individual response vector, we can calculate the continuous phenotype as: $\boldsymbol{y}_i = \boldsymbol{x}_i\boldsymbol{\beta}_k + \boldsymbol{e}_i$, where $\boldsymbol{e}_i \sim \mathcal{N}_q(\boldsymbol{0}, \boldsymbol{\Sigma}_e)$ and $\boldsymbol{\Sigma}_e$ is an equicorrelation matrix whose diagonal entries are equal to 1 and off-diagonal entries are $\rho_e \sim \mathcal{U}(0.0, 0.2)$. Finally, the mixed-type outcomes can be generated by composing 50% continuous and 50% binary variables, where the binary variables were obtained by binarizing continuous responses based on its median value.
  5. [Effect size] We assumed that causal variants are associated with half or all of phenotype outcomes, denoted by $\gamma = 50\%, 100\%$. We considered arbitrary 16 genetic regions where each has 5 genetic variants and the 5 genetic variants can have different sizes of association with a single phenotype, as well as a variant can have different sizes of association across multiple phenotypes. Consequently, for $v \in \{1, \cdots, \gamma q\}$, the vector of effect sizes $(\beta_{u,v}, \beta_{u+1,v}, \cdots, \beta_{u+4,v})$ was set as

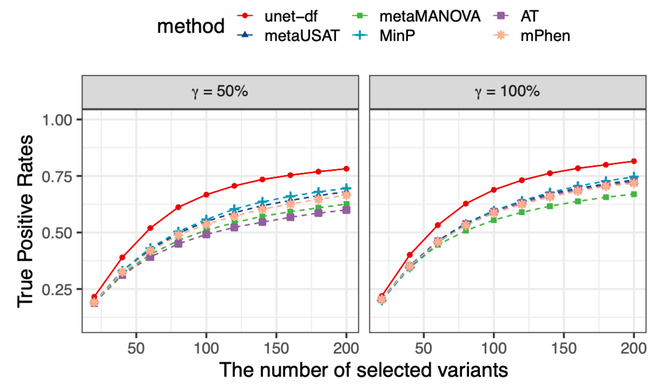
$$(0.5, 0.75, 1.0, 1.25, 1.5) \quad \text{for } u = 100, 200, \cdots, 800, \tag{4}$$

and for $u = 900, 200, \cdots, 1600$

$$\begin{array}{ll} (0.5, 0.75, 1.0, 1.25, 1.5) & \text{if } v \text{ is an even number,} \\ (1.5, 1.25, 1.0, 0.75, 0.5) & \text{if } v \text{ is an odd number.} \end{array} \tag{5}$$

Otherwise $\beta_{u,v} = 0$. The sample size, the number of variants and the number of phenotypes are fixed at $n = 400$, $p = 40000$ and $q = 8$, respectively, to mimic our real data as as much as possible.

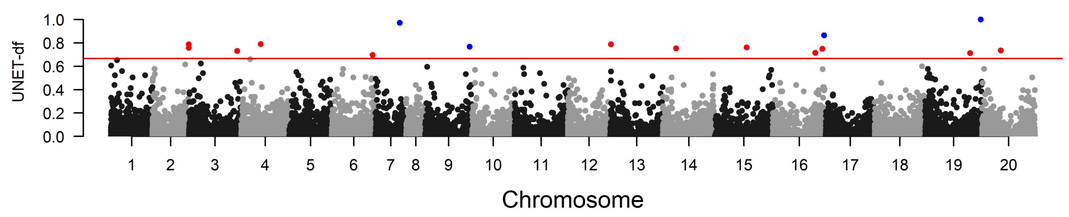
  6. [Comparing Methods] The existing individual methods: "MinP" (Am. J. Hum. Genet. 2007), "MultiPhen" (PLOS one. 2012), "MetaUSAT" (Genet. Epidemiol. 2018) and "AT" (Bioinformatics. 2019), among which MultiPhen only uses individual-level data set.
  7. [Evaluation] To evaluate our proposed method, we compared the true positive rates (TPR) of the proposed method with the comparing methods. We first calculated the selection probability of the proposed method and p-values of the comparing methods for all SNPs. Next, and then we calculated the TPRs based on the top-ranked SNPs by selection probability and p-valus. We finally compared the average of TPRs after repeating this procedure 100 times.



Real data analysis

- We analyzed **peanut dataset** composed of 301 individuals and 32,145 SNPS with 5 continuous outcomes such as seed aspect ratio, chlorophyll, flowering date, etc. They are also weakly correlated with each other similarly as in simulation studies.
- We calculated selection probabilities of the elastic-net model for each phenotype and the proposed "UNET-df", and we also computed the **empirical thresholds** for the selection probabilities similarly as Kim et al., 2020.
- **Manhattan plot** displays the selection probabilities along with 20 chromosomes with red lines representing the empirical thresholds, and it shows that a total number of 17 SNPs out of 32,145 was selected by our proposed method "UNET-df".
- **The following table** lists the results on the top 10 SNPs by selection probability of "UNET-df" due to space limit.
- Among them, the 10th SNP "AX-176815650" was selected as top 10 by "UNET-df" but was not selected by none of the elastic-net models. However, the elastic-net models have relatively high selection probabilities, which means that our proposed method can find out potentially outcome-related SNP by incorporating multilpe associations with weak signal.

SNP	ENET					UNET-df
	SAR	LC	FD	LAR	SC	
AX-147262340	0 (4645)	0 (5944)	0.74 (1)*	0.44 (16)*	0.01 (4140)	1 (1)*
AX-177639817	0 (4645)	0.04 (1199)	0.38 (28)	0.01 (4202)	0.53 (8)*	0.97 (2)*
AX-176800793	0 (4645)	0 (5944)	0.25 (104)	0.6 (4)*	0 (7318)	0.86 (3)*
AX-176797194	0 (4645)	0.01 (3148)	0 (6878)	0.2 (163)	0.61 (2)*	0.79 (4)*
AX-147243026	0 (4645)	0.11 (327)	0.64 (4)*	0 (7547)	0.06 (1020)	0.79 (5)*
AX-147243027	0 (4645)	0.11 (327)	0.64 (4)*	0 (7547)	0.06 (1020)	0.79 (5)*
AX-176802856	0 (4645)	0.11 (327)	0.64 (4)*	0 (7547)	0.06 (1020)	0.79 (5)*
AX-176807953	0 (4645)	0.03 (1539)	0.61 (8)*	0.07 (920)	0.01 (4140)	0.77 (8)*
AX-176803005	0 (4645)	0 (5944)	0.11 (497)	0.68 (2)*	0.06 (1020)	0.76 (9)*
AX-176815650	0 (4645)	0.29 (42)	0.3 (52)	0 (7547)	0.03 (2044)	0.76 (10)*



Conclusions

- In real life, there are many types of phenotypes, and many are eager to detect the pleiotropic variants associated with them.
- In this work, we proposed new variable selection strategy to identify pleiotropic variants by unifying the regularization models using selection probabilities even when the outcome-related variants have different sizes of assoications across multiple phenotypes.
- Compared to the existing meta-analysis methods, it is more powerful even when half of phenotypes is associated with the outcome-related variants. Furthermore, it provides more stable selection results regardless on tuning parameters by using resampling approach.
- We successfully identified several candidate pleiotropic variants associated with multiple phenotype outcomes in the peanut datasets.
- However, our proposed method can only investigate whether or not there is any significant association between a specific variant and multiple phenotypes. Accordingly, the post-hoc selection would be an interesting problem for future research.